

BACKGROUND & MOTIVATION

- Almost **three-quarters of a million** Canadians currently live with dementia — not normal aging, but cognitive and functional trajectories that diverge from norms **over time**.
- Prevention aims to **delay onset** and **slow progression** by targeting **modifiable risk factors (MRFs)**.
- Son et al.¹ estimate that **12 modifiable risk factors** may account for **≈ 49.2%** of dementia cases in Canada.

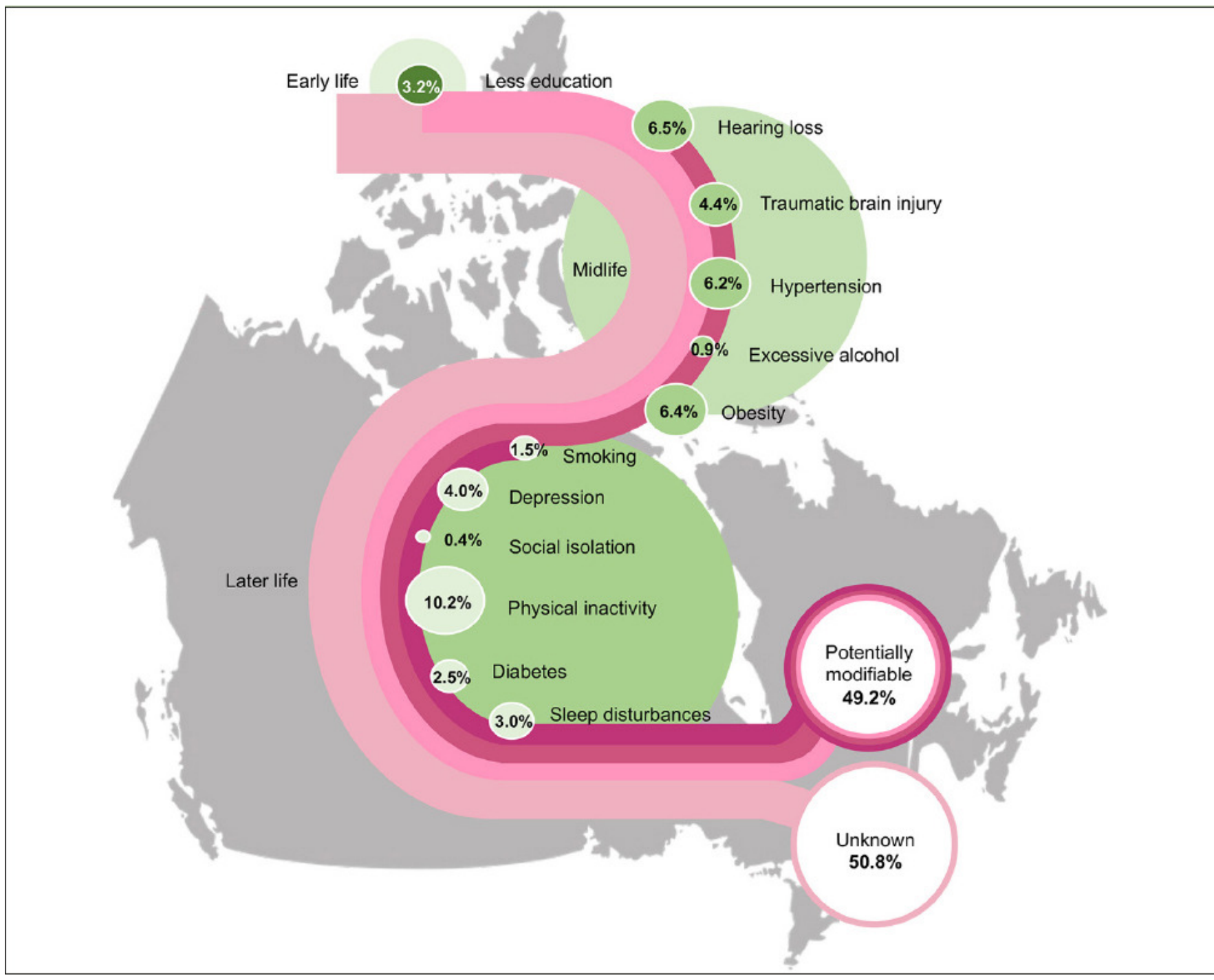


Figure 1. Population-attributable fractions of dementia MRFs in Canada¹.

Goal. Quantify how changing actionable risk factors shifts heterogeneous cognitive trajectories.

CAUSAL QUESTION & ESTIMAND

If we intervene on a **modifiable risk factor**, how does the distribution of **latent trajectory patterns** change?

Let ζ_i^a be the trajectory class subject i would follow under risk-factor level $a \in \{0, 1\}$:

$$\psi_k(a) = P(\zeta_i^a = k), \quad \text{ATE}_k = \psi_k(1) - \psi_k(0)$$

Two features make this hard:

- Cognitive status is latent & multivariate-longitudinal** — multiple tests that jointly define trajectory patterns.
- The number of patterns K is unknown.** Conventional latent-class methods have difficulty propagating uncertainty into the causal estimand.

⇒ motivates a **full Bayesian nonparametric framework**.

CAUSAL MODEL & ASSUMPTIONS

Setup

- L_i : baseline confounders (demographics + modifiable factors).
- $A_i \in \{0, 1\}$: modifiable risk factor (exposure).
- ζ_i : latent cognitive-trajectory class.
- Y_{ij} : multivariate cognitive outcomes at visit $j = 1, \dots, J$; manifest indicators of ζ_i .

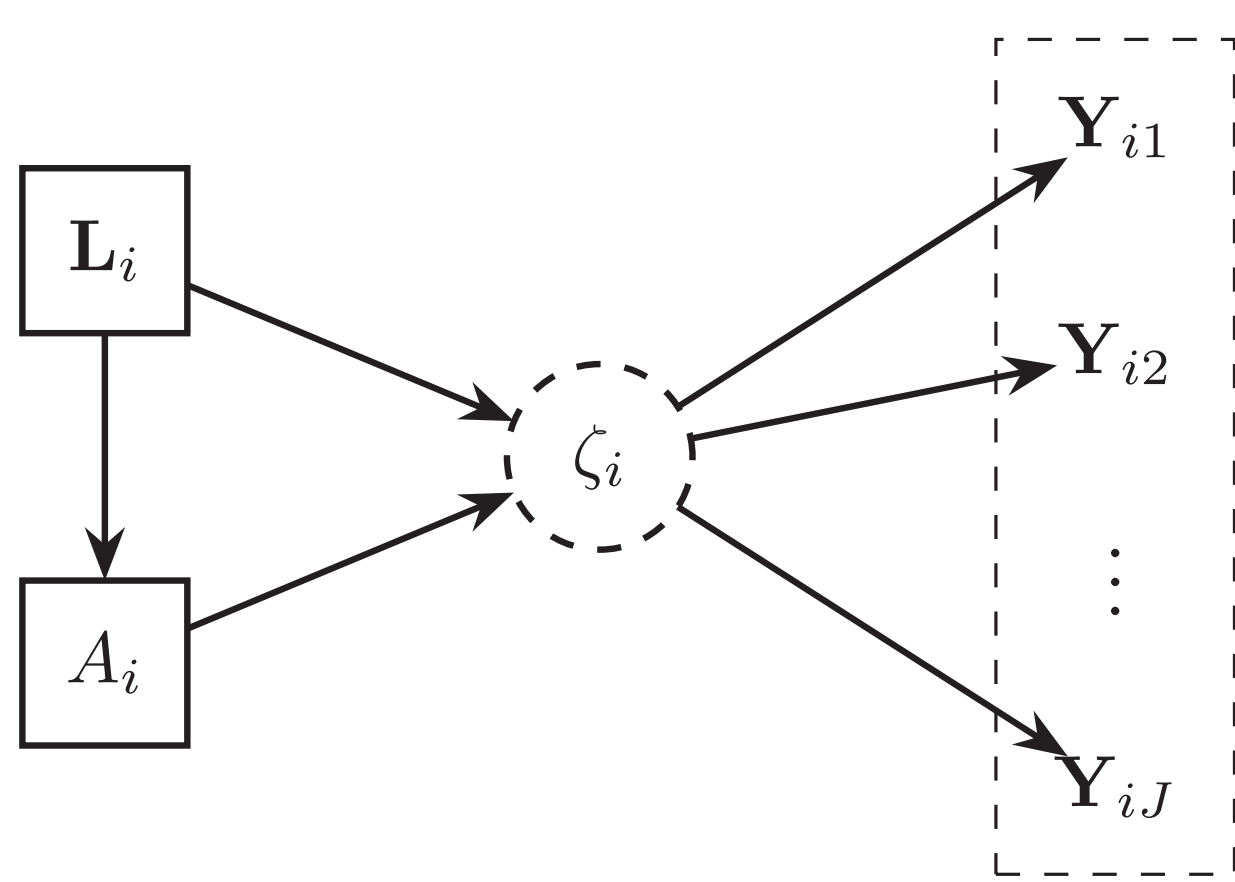


Figure 2. Causal DAG: baseline confounders & treatment → latent class → multivariate longitudinal outcomes.

Identifying assumptions

- SUTVA and Positivity.**
- Latent exchangeability:** $A_i \perp\!\!\!\perp \zeta_i^a \mid L_i$.
- Conditional independence:** $Y_i \perp\!\!\!\perp (A_i, L_i) \mid \zeta_i$.

BNP LATENT CLASS MODEL

1. From finite to infinite mixtures

A traditional latent-class model fixes K :

$$f(\mathbf{Y}_i \mid X_i, Z_i, h_i) = \sum_{k=1}^K \pi_k(h_i) \phi(\mathbf{Y}_i; \mu_{ik}, \Sigma_{ik}).$$

Koo & Kim² replace it with an **infinite mixture** using a large truncation K and a **predictor-dependent normalized gamma-process prior**:

$$\pi_k(h_i) = \frac{w_k \exp(h_i^\top \gamma_k)}{\sum_{\ell=1}^K w_\ell \exp(h_i^\top \gamma_\ell)}, \quad w_k \sim \text{Gamma}(\alpha/K, 1).$$

- w_k : baseline weight for class k ; γ_k : how predictors shift allocation; α : tendency to occupy more classes.
- With $\gamma_k = 0$, $\pi_k = w_k / \sum_{\ell} w_\ell$ — a finite approximation to a **Dirichlet-process mixture**.

2. Class-specific longitudinal outcome

$$\mathbf{Y}_i \mid \zeta_i = k = X_i \beta_k + Z_i b_i + \epsilon_i, b_i \sim \text{MVN}(0, \Lambda_k), \epsilon_i \sim \text{MVN}(0, \sigma^2)$$

JOINT LIKELIHOOD & BAYES G-COMP

1. Joint likelihood of the observed data

$$L(\Lambda) \propto \prod_{i=1}^n p(L_i \mid \Theta_L) p(A_i \mid L_i, \Theta_A) \times \sum_{k=1}^{\infty} p(\mathbf{Y}_i^{\text{obs}} \mid \zeta_i = k, \Theta_Y) p(\zeta_i = k \mid A_i, L_i, \Theta_\zeta)$$

with $\Lambda = \{\Theta_Y, \Theta_L, \Theta_A, \Theta_\zeta\}$.

2. Causal estimand (average potential outcome)

$$E[\psi_k(a) \mid O_{1:n}] = \int p(\zeta_i = k \mid A_i = a, L^*, \Theta_\zeta) \times p(L^* \mid \Theta_L) p(\Lambda \mid O_{1:n}) dL^* d\Lambda.$$

Estimated by **parametric Bayesian g -computation**³; fit in **Stan** via cmdstanr:

- Fit the joint likelihood with non-informative priors.
- Draw posterior samples $\Lambda^{(s)}$.
- Simulate synthetic covariates $L^{(s,r)} \sim p(\cdot \mid \Theta_L^{(s)})$.
- Posterior-predictive ATE:

$$\widehat{\text{ATE}}_k = \frac{1}{SM} \sum_{s,r} [P(\zeta = k \mid A=1, L^{(s,r)}, \Theta_\zeta^{(s)}) - P(\zeta = k \mid A=0, L^{(s,r)}, \Theta_\zeta^{(s)})]$$

PRACTICAL NOTE: LABEL SWITCHING

- Mixture posteriors are **symmetric in class labels**, so chains freely permute class indices across draws.
- Fix:** per draw, solve the linear sum assignment problem (**Hungarian matching**) against a fixed pivot; apply the same permutation to $B_k, \log w_k, \gamma_k$ before pooling for g -computation.

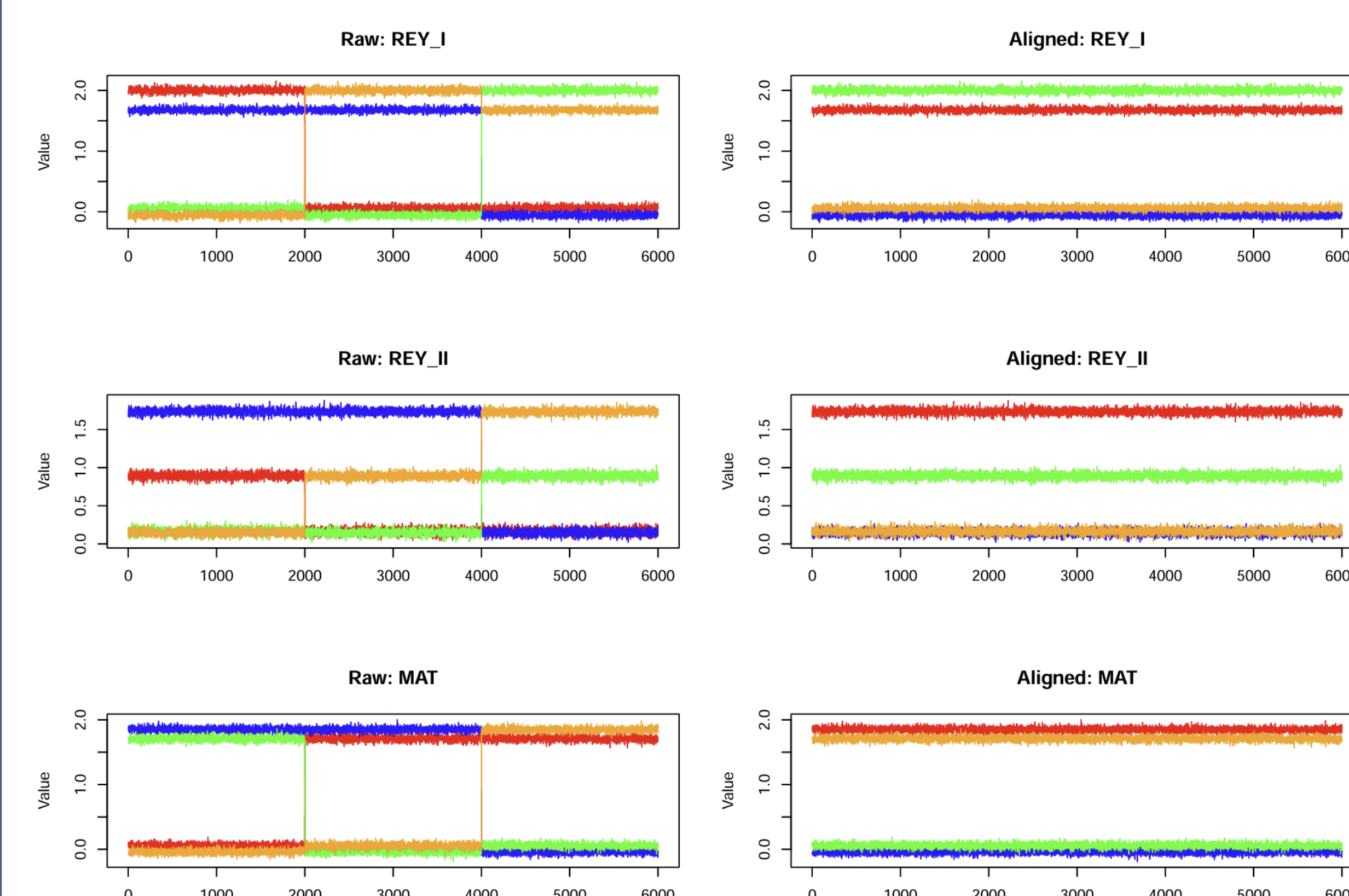


Figure 3. Trace plots of class-specific outcome means before (left) and after (right).

Unaligned labels inflate posterior variance and bias ATE_k toward 0; alignment is required before causal estimation.

SIMULATION STUDY

Design — $n=300$, $J=6$ visits, $p=3$ cognitive outcomes, $K_{\text{true}}=4$ classes (class-specific cubic-polynomial random-intercept model); binary treatment ($\sim 30\%$ prevalence); 100 replications/scenario on HPC (SLURM).

- Proposed:** BNP-LCM in Stan ($K_{\text{max}}=6$; 3 chains).
- Baseline:** IPW-weighted FlexMix, BIC-selected $K \in \{2, \dots, 6\}$, bootstrap CIs.

Simulation result (across 100 reps; RMSE & coverage on K -correct reps):

Scenario	Proposed (BNP-LCM)			Baseline (IPW-FlexMix)		
	K -corr. %	RMSE	Cov. %	K -corr. %	RMSE	Cov. %
Baseline	89	0.052	95.8	62	0.083	95.6
Null ATE	85	0.053	95.0	52	0.059	97.6
Imbalance	79	0.050	94.9	56	0.081	96.9
Large n	90	0.035	95.6	60	0.068	95.8

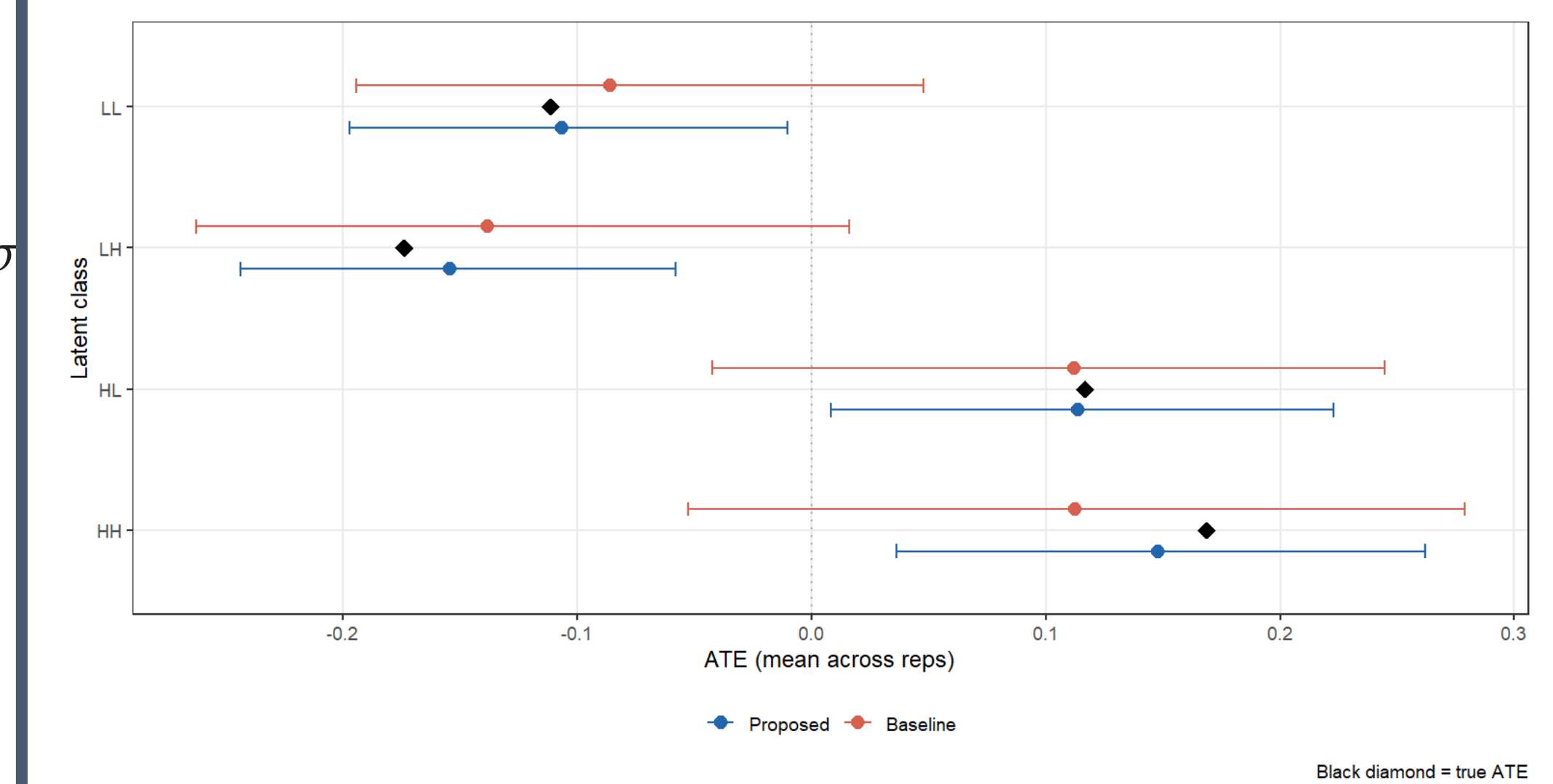


Figure 4. Per-class ATE (baseline scenario, K -correct reps). ♦ = true ATE.

The proposed BNP-LCM attains **higher K -recovery** and **lower RMSE** across all scenarios, with **near-nominal coverage** and **tighter intervals** via Bayesian g -computation.

APPLICATION: CANADIAN LONGITUDINAL

Cohort ($n=2,694$; age ≥ 65 , complete data over **3 waves**)⁵.

- Exposure A_i :** baseline depression (CES-D-10 ≥ 10).
- Outcomes Y_{ij} :** 6 standardized tests — memory (REY_I, REY_II) & executive function (MAT, AF1, FAS, STP).
- Confounders L_i :** demographics + modifiable factors¹; truncation $K_{\text{max}}=10$.

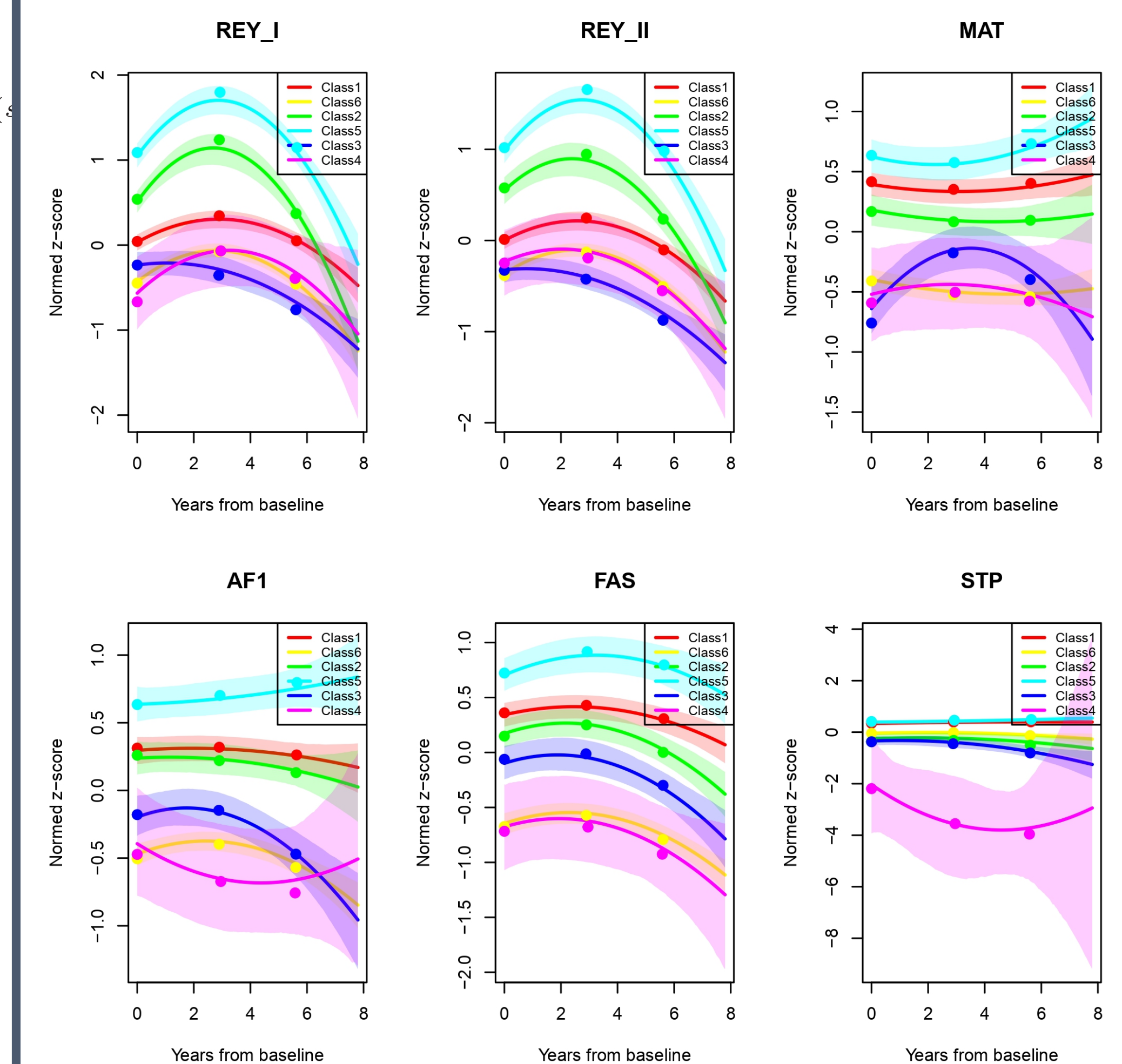


Figure 5. 6 active trajectory classes recovered; posterior-mean normalized z -scores with 95% bands. Classes differ by level and domain-specific change.

Class	Descriptive interpretation	Mean	2.5%	97.5%
1	Favourable exec. fn., avg. memory	-0.129	-0.191	-0.064
2	Memory-preserved, weaker Stroop	0.021	-0.038	0.085
3	Lower traj., memory decline, high depr.	0.093	0.031	0.160
4	Very small severe-impairment traj.	0.004	-0.007	0.024
5	High cognitive-function traj.	-0.024	-0.073	0.029
6	Lower memory & executive-fn. traj.	0.035	-0.038	0.113

Depression shifts the population away from favourable and toward lower-performing cognitive-aging trajectories.